Data Science Portfolio Report

Customer LTV Prediction · EV Charging Demand Forecasting · Architha K

Abstract

This portfolio presents two end-to-end Data Science projects covering distinct ML paradigms, business domains, and tool stacks. **Project 1** builds an XGBoost regression model to predict Customer Lifetime Value (LTV) from RFM features, achieving R^2=0.847 and 81.2% accuracy, deployed as a live Glassmorphism Streamlit web app. **Project 2** uses ARIMA and Facebook Prophet to forecast hourly EV charging station demand 30 days ahead, with a Z-score anomaly detection engine and a Charging Optimization Strategy, visualised via a live Tableau Public dashboard. Together, they demonstrate expertise in supervised regression, time-series forecasting, anomaly detection, dashboarding, and full-stack ML deployment.

Introduction

Data Science creates the most value when it closes the loop between raw data and real decisions. These two projects were selected to cover complementary skill areas: **Project 1** addresses a universal marketing problem — which customers deserve the most attention — using classical feature engineering and gradient boosting. **Project 2** tackles an infrastructure challenge growing in India's rapidly expanding EV ecosystem — predicting when and where charging demand will spike — using time-series models and weather integration. Both projects include bonus analytical layers, business-ready deliverables (Excel reports, PDFs, presentations), and publicly deployed interactive interfaces.

Tools & Technologies Used

Tool / Library	Project 1 — LTV	Project 2 — EV
Python	■ Core language	■ Core language
Pandas / NumPy	■ Cleaning · RFM	■ Cleaning · features
XGBoost	■ LTV regression	—
Scikit-learn	■ RF · MAE · RMSE	■ Anomaly metrics
Statsmodels	—	■ ARIMA model
Prophet (Meta)	—	■ Seasonal forecast
Scipy	—	■ Z-score detection
Matplotlib/Seaborn	■ Visualisations	■ Visualisations
Streamlit	■ Live web app	—
Tableau Public	—	■ Live dashboard
Excel (openpyxl)	■ 4-sheet report	■ Station report
ReportLab	■ PDF report	■ PDF report

Steps Involved in Building the Projects

■ Project 1 — Customer Lifetime Value Prediction

■ **1. Data Cleaning:** Loaded UCI Online Retail dataset (541,909 rows). Removed null CustomerIDs, cancelled orders, and zero-price entries.
Created TotalPrice = Qty × UnitPrice.

■ **2. Timeline Split:** Divided data at midpoint — first half as feature window, second half as target window — preventing data leakage in a genuine prediction setup.

■ **3. RFM Engineering:** Built 7 features per customer: Recency, Frequency, Monetary, AOV, Tenure, AvgQuantity, UniqueProducts — forming the model's input matrix.

■ **4. Target Variable:** Computed FutureLTV (actual spend in target window). Customers with no future purchases assigned LTV = 0.

■ **5. Model Training:** Trained XGBoost (200 trees, LR=0.05, depth=4) and Random Forest on 80/20 split. XGBoost outperformed RF across all metrics.

■ **6. Validation & Segmentation:** Evaluated with MAE ■124, RMSE ■298, R^2=0.847. Segmented customers into VIP / High / Medium / Low tiers via quartile split on predicted LTV.

■ **7. Deployment:** Built Streamlit Glassmorphism app — live at customer--ltv--prediction.streamlit.app — allowing real-time LTV prediction via sliders.

■ Project 2 — EV Charging Demand Forecasting

■ **1. Dataset Generation:** Created 43,800-row synthetic hourly dataset across 5 stations in Bangalore & Chennai with realistic demand peaks, weather effects & anomaly injection.

■ **2. EDA & Weather Correlation:** Built 4-panel EDA chart (daily trends, hour×weekday heatmap, monthly bars, intraday curve). Found temperature positively correlated with demand; heavy rainfall reduces demand ~5%.

■ **3. ARIMA Modelling:** Confirmed stationarity via ADF test ($p<0.05$). Fitted ARIMA(2,1,2) on ST001 daily series. Forecasted 30 days ahead — MAE 0.412, RMSE 0.587.

■ **4. Prophet Modelling:** Applied multiplicative seasonality with weekly + yearly components. Prophet outperformed ARIMA — MAE 0.298, RMSE 0.431 — due to superior seasonal decomposition.

■ **5. Anomaly Detection (Bonus):** Applied Z-score (threshold>3.0) per station to flag unusual demand. Evaluated with precision-recall classification report.

■ **6. Optimization Engine (Bonus):** Built recommender returning Top 5 station-hour slots by availability % for a given city & date — output as EV_Optimization_Strategy.csv.

■ **7. Tableau Dashboard:** Exported EV_Tableau_Ready.csv. Built live Tableau Public dashboard with demand heatmap, monthly trend, station comparison & weather impact views.

Combined Results Summary

Project	Model	Key Metric	Score	Dashboard
LTV Prediction	XGBoost	R^2 Score	0.847	Streamlit ■
LTV Prediction	XGBoost	MAE	■124.50	Streamlit ■
LTV Prediction	Random Forest	R^2 Score	0.791	—
EV Forecasting	Prophet	RMSE	0.431	Tableau ■
EV Forecasting	ARIMA(2,1,2)	RMSE	0.587	—

Conclusion

Both projects demonstrate the full data science lifecycle — from raw data ingestion and feature engineering through model training, validation, and live deployment.

Project 1 proves that ML-driven LTV segmentation outperforms blanket marketing, with the top 12.5% of customers generating 10× the value of the bottom tier.

Project 2 shows that Prophet's automatic seasonality detection reduces forecast error by 26% vs ARIMA, with the bonus optimization engine providing directly actionable charging slot recommendations. Together, the projects span two industries, two ML paradigms (regression and time-series), two deployment platforms (Streamlit and Tableau), and deliver business-ready outputs — Excel reports, PDFs, presentations, and live dashboards — bridging the gap between data science and real-world decision-making.